

Pick a Platter

Some of the most dramatic decreases in prices of computer hardware have been that of hard disk drives. You can now get an entry-level 20 GB IDE hard disk for as little as Rs 4,600, which is just 25 paise per MB! The entry-level 5400 rpm drives deliver sequential read/write speeds of about 25MBps which is sufficient for applications like word processors, Internet surfing, etc. These drives are at least ATA/66 compliant (which implies a peak theoretical transfer rate of 66 MBps).

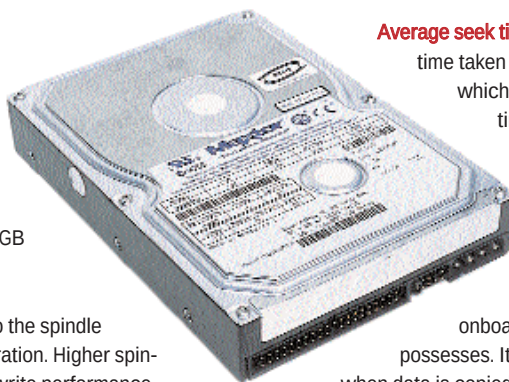
The higher performance 7200 rpm IDE models are generally ATA/100 compliant and are ideally suited to applications like audio and video processing. The prices for these drives begin from Rs 6,800 for a 20 GB unit. When it comes to sheer performance, however, SCSI still takes the honours with Ultra160 SCSI—a standard that promises a theoretical transfer rate of 160MBps. However, this technology comes at a price.

SMART TIPS

Capacity: For most users an entry-level 20 GB drive would be sufficient. However, IDE drives of capacity up to 100 GB are available. SCSI models are available in sizes ranging from 4.3 GB to 80 GB depending on the drive rpm.

Rotational speed : This refers to the spindle rpm of the drive under consideration. Higher spindle speeds lead to faster read/write performance. These range from 5400 to 7200 rpm for IDE drives and 7200 to 15000 rpm for SCSI models.

Sustained data transfer rates: This is a measure of the drive's performance in terms of data transfer to and from the drive. This should be about 20-30 MBps for 5400-rpm drives and 30-50 MBps for 7200-rpm drives.



Average seek time: This is the average time taken by the drive to access data which is stored on the drive. A time that is between 5.5 to 10 seconds is acceptable for most mainstream applications.

Data buffer (cache): This is the amount of onboard memory that the drive possesses. It increases transfer speeds when data is copied from one location in the drive to another. This ranges from 512 KB in the 5400-rpm drives to as much as 2 MB in the 7200-rpm IDE models.

Motherboard/IDE controller card: In order to tap the full potential of a hard drive, you need to make sure that your motherboard's IDE controller supports the same specification (ATA-66/100) interface.

Also Remember...

■ Your drives performance will see a major improvement (up to 25-30 per cent) if Direct Memory Access (DMA) is enabled in your system.

■ If you are buying a SCSI drive, you will need to buy an appropriate SCSI adapter depending on the SCSI standard that the drive uses.

■ Most new drives incorporate some form of noise reduction technology. This will impact drive performance by about 5-7 per cent but makes for much quieter operation.

■ If you have an ATA/100 or a SCSI drive, make sure that you have the correct cables to connect your hard disk to the system. For example, an ATA/100 hard disk drive will work if you use a standard 40-pin, 40-conductor cable. However, for it to function as an ATA/100 drive, you will need a 40-pin 80-conductor cable.

	General home use	Performance-hungry home and office applications	Specialist high-bandwidth applications
You need	A hard disk for general applications such as word processing and Internet surfing	Larger amounts of data transfer as you're running more demanding applications like image editing and animation	A hard disk that will support applications that demand the highest in performance and reliability like video capture.
Look for	20 GB minimum at 5400 rpm, supporting ATA-66 transfer rates. Go for the largest capacity that fits your budget	30 GB minimum at 7200 rpm, supporting ATA-100 transfer rates. Go for the largest capacity that fits your budget	Minimum 10000-rpm drive, supporting Ultra2 SCSI or better transfer rates. Choose capacity according to your application.
Our pick	Seagate U10 20GB	Seagate Barrucudda ATA-III/IV, IBM Deskstar 75GXP	Seagate Cheetah SCSI Ultra160
Price range	Up to Rs 6,500	Up to Rs 10,000	Rs 15,000 to Rs 40,000

